

COMPUTATIONAL TECHNIQUES IN CIVIL ENGINEERING (Water Resources Part)

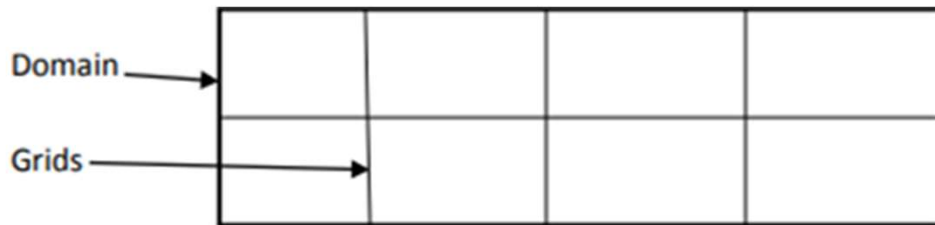
Internal Evaluation (20 marks)

1. Structure Part -10 marks
2. Water Resources Part -10 marks
 - i. Class Attendance – 2 marks
 - ii. Assignment – 3 marks
 - iii. Examination/Test – 5 marks

5. Finite Difference Method (FDM)

5.1 Introduction

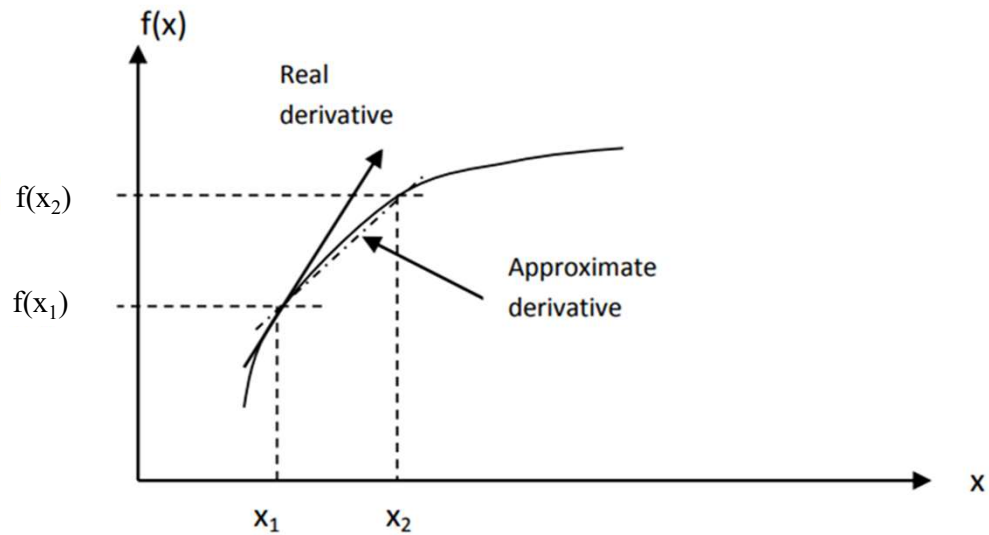
Finite difference method is a numerical technique of solving differential equation (DE) in which the domain is represented by an array of regularly spaced grid points (rows and columns), and the DE is converted to difference equation at a grid point. Hence, the method approximates the governing equation point wise. The value of a function at a point is related to value of nearby points. The method is simple and easy to implement, but it is difficult to apply for irregular geometry.



Mathematical concept of FDM

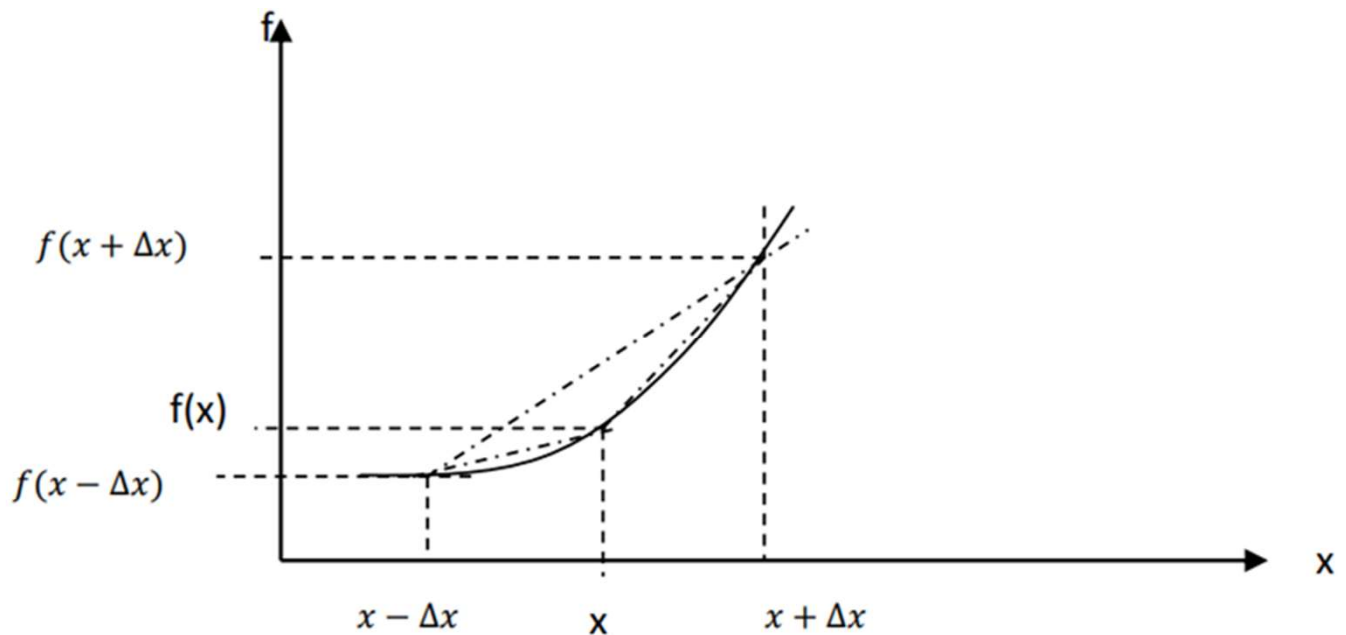
- Mathematically, DE is replaced them with difference equations over a small step in FDM.
- Real derivative is represented by tangent to a point on a curve. In FDM, the derivative is evaluated approximately by taking the slope of line joining two points.

$$\frac{df}{dx} = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$



- Finite difference equations (FDE) are obtained by using Taylor's series approximation to DE.

Finite difference approximation of first derivative, df/dx



a. Forward difference approximation

Taylor's series expansion of $f(x)$ at $x+\Delta x$ is

$$f(x + \Delta x) = f(x) + \Delta x \frac{\partial f}{\partial x} + \frac{\Delta x^2}{2!} \frac{\partial^2 f}{\partial x^2} + \frac{\Delta x^3}{3!} \frac{\partial^3 f}{\partial x^3} + \dots \dots \quad (I)$$

Neglecting second and higher order terms

$$\frac{\partial f}{\partial x} = \frac{f(x+\Delta x) - f(x)}{\Delta x}$$

Dropping the higher order terms is known as truncation.

b. Backward difference approximation

Taylor's series expansion of $f(x)$ at $x-\Delta x$ is

$$f(x - \Delta x) = f(x) - \Delta x \frac{\partial f}{\partial x} + \frac{\Delta x^2}{2!} \frac{\partial^2 f}{\partial x^2} - \frac{\Delta x^3}{3!} \frac{\partial^3 f}{\partial x^3} + \dots \dots \quad (II)$$

Neglecting second and higher order terms

$$\frac{\partial f}{\partial x} = \frac{f(x) - f(x-\Delta x)}{\Delta x}$$

c. Central difference approximation

Subtracting eq. (II) from eq. (I), and neglecting third and higher order terms

$$f(x + \Delta x) - f(x - \Delta x) = 2\Delta x \frac{\partial f}{\partial x} \quad (III)$$

$$\frac{\partial f}{\partial x} = \frac{f(x+\Delta x) - f(x-\Delta x)}{2\Delta x}$$

Finite difference approximation of second derivative

Adding eq. (I) and eq. (II), solving for and neglecting higher order terms

$$\frac{\partial^2 f}{\partial x^2} = \frac{f(x+\Delta x) - 2f(x) + f(x-\Delta x)}{\Delta x^2}$$

Steps in FDM

- I. Discretization of domain: Replacing the continuous spatial and temporal domain by grid points or cells and time levels
- II. Discretization of PDE: Replacing PDE by a set of algebraic equations known as Finite difference equations (FDEs)
- III. Specification of solution algorithm: Step by step procedure for solving FDE at each point/grid

Order of accuracy of schemes

Order of accuracy represents how well the solution of finite difference method approximates the real solution of differential equations.

Considering the forward scheme, From eq. (I)

$$f(x + \Delta x) = f(x) + \Delta x \frac{\partial f}{\partial x} + \frac{\Delta x^2}{2!} \frac{\partial^2 f}{\partial x^2} + \frac{\Delta x^3}{3!} \frac{\partial^3 f}{\partial x^3} + \dots$$

Neglecting second and higher order terms

$$\frac{f(x+\Delta x) - f(x)}{\Delta x} - \frac{\partial f}{\partial x} = \frac{\Delta x}{2} \frac{\partial^2 f}{\partial x^2} + \frac{\Delta x^2}{6} \frac{\partial^3 f}{\partial x^3} + \dots$$

RHS = truncation error

The lowest power of Δx in the truncation error determines the order of accuracy. This scheme is first order accurate. As Δx tends to zero, the error in the higher order scheme tends to zero more rapidly than lower order scheme.

Similarly, backward scheme is also first order accurate.

Considering central difference scheme, Subtracting eq. (II) from eq. (I)

$$f(x + \Delta x) - f(x - \Delta x) = 2\Delta x \frac{\partial f}{\partial x} + \frac{\Delta x^3}{3} \frac{\partial^3 f}{\partial x^3} + \dots$$

Dividing by $2\Delta x$ and rearranging

$$\frac{f(x+\Delta x) - f(x-\Delta x)}{2\Delta x} - \frac{\partial f}{\partial x} = \frac{\Delta x^2}{6} \frac{\partial^3 f}{\partial x^3} + \dots$$

This is second order accurate scheme.

Concept of Consistency, convergence and stability

Consistency: If continuous PDEs (real) are obtained from FDEs as Δx and Δt approaches to zero, then the resulting solution is said to be consistent. Hence, consistency is related to equations. Taylor's series expansion is applicable for consistency analysis.

Consistency test: example $\frac{\partial Q}{\partial x} + \frac{\partial A}{\partial t} = 0$

Considering forward difference scheme

$$\frac{Q(x+\Delta x)-Q(x)}{\Delta x} + \frac{A(t+\Delta t)-A(t)}{\Delta t} = 0 : \text{Finite difference equation}$$

Taylor's series expansion

$$Q(x + \Delta x) = Q(x) + \Delta x \frac{\partial Q}{\partial x} + \frac{\Delta x^2}{2!} \frac{\partial^2 Q}{\partial x^2} + \frac{\Delta x^3}{3!} \frac{\partial^3 Q}{\partial x^3} + \dots$$

$$\frac{Q(x+\Delta x)-Q(x)}{\Delta x} - \frac{\partial Q}{\partial x} = \frac{\Delta x}{2} \frac{\partial^2 Q}{\partial x^2} + \frac{\Delta x^2}{6} \frac{\partial^3 Q}{\partial x^3} + \dots$$

$$\text{As } \Delta x \rightarrow 0, \frac{Q(x+\Delta x)-Q(x)}{\Delta x} = \frac{\partial Q}{\partial x}$$

$$A(t + \Delta t) = A(t) + \Delta t \frac{\partial A}{\partial t} + \frac{\Delta t^2}{2!} \frac{\partial^2 A}{\partial t^2} + \frac{\Delta t^3}{3!} \frac{\partial^3 A}{\partial t^3} + \dots$$

$$\text{Similarly, as } \Delta t \rightarrow 0, \frac{A(t+\Delta t)-A(t)}{\Delta t} = \frac{\partial A}{\partial t}$$

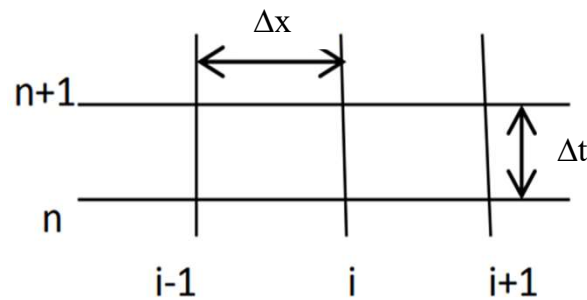
Hence the real PDE is obtained and the numerical scheme is consistent.

Convergence: If solution of FDM approaches the true solution as Δx and Δt approaches to zero, then the resulting solution is said to be convergent. Hence, convergence is related to the solution of equations. Comparison of analytical solution if exists with numerical solution provides a way to see the convergence.

Stability: If the round-off and truncation errors do not accumulate to cause the solution to diverge, then the solution is said to be stable. Fourier analysis is performed for stability analysis.

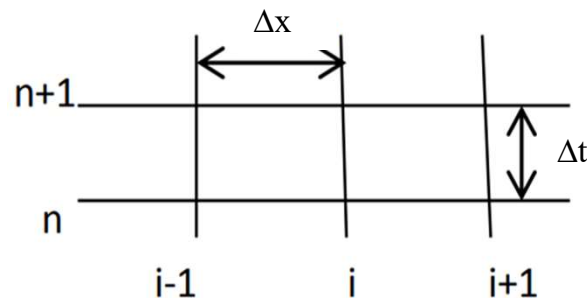
5.2 Explicit and Implicit Schemes

Explicit Scheme: If the solution at $t = n+1$ is obtained from the conditions at $t = n$, the scheme is known as explicit scheme. In such scheme, the equations are solved sequentially. The explicit scheme is simpler to implement, but it can be unstable. It is also convenient because results are given at grid points and it can treat slightly varying channel geometry from section to section. It is less efficient than implicit method and hence not suitable for simulation over long time period.



Implicit Scheme: If the solution at $t = n+1$ is obtained from the conditions at $t = n$ and $n+1$, the scheme is known as implicit scheme. In such scheme, the equations are solved simultaneously. Implicit scheme is mathematically more complicated. The method is stable for large computation steps and hence works much faster than explicit scheme. The method can handle channel geometry varying significantly from one section to the next.

Example of Discretization in space domain



Explicit FDEs: Use time line n (spatial derivatives evaluated by using known values)

$$\text{Forward: } \frac{\partial f}{\partial x} = \frac{f_{i+1}^n - f_i^n}{\Delta X}, \text{ Backward: } \frac{\partial f}{\partial x} = \frac{f_i^n - f_{i-1}^n}{\Delta X}, \text{ Central: } \frac{\partial f}{\partial x} = \frac{f_{i+1}^n - f_{i-1}^n}{2\Delta X}$$

Implicit FDEs: Use time line n+1

$$\text{Forward: } \frac{\partial f}{\partial x} = \frac{f_{i+1}^{n+1} - f_i^{n+1}}{\Delta X}, \text{ Backward: } \frac{\partial f}{\partial x} = \frac{f_i^{n+1} - f_{i-1}^{n+1}}{\Delta X}, \text{ Central: } \frac{\partial f}{\partial x} = \frac{f_{i+1}^{n+1} - f_{i-1}^{n+1}}{2\Delta X}$$

Second order derivative: Example of discretization

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) = \frac{1}{\Delta x} \left[\left(\frac{\partial f}{\partial x} \right)_{i+1}^n - \left(\frac{\partial f}{\partial x} \right)_i^n \right] = \frac{1}{\Delta x} \left[\frac{f_{i+1}^n - f_i^n}{\Delta X} - \frac{f_i^n - f_{i-1}^n}{\Delta X} \right] = \frac{1}{\Delta x^2} (f_{i+1}^n - 2f_i^n + f_{i-1}^n)$$

5.3 Unsteady non-uniform flow equation in open channel: Saint Venant Equations

The equations of unsteady non-uniform flow in 1D is known as Saint Venant Equations. These equations are based on continuity and momentum principle.

Example of occurrence of unsteady flow in open channel

- ❖ Flood waves in rivers, streams and drainage channels due to rainfall, due to failure of dams/dykes,
- ❖ Storm water flow in sewers or drainage channels,
- ❖ Surge in power canals or tunnels due to opening/closing of control gates or starting/stopping of turbines, pumps

Assumptions in Saint Venant Equations

- The channel is prismatic.
- The flow is 1D, i.e. the velocity is uniform in a cross-section.
- Flow is assumed to vary gradually along the channel so that hydrostatic pressure prevails and vertical accelerations can be neglected.
- The channel is approximated as a straight line.
- The bottom slope of the channel is small and the channel bed is fixed.
- Resistance coefficients for steady uniform turbulent flow are applicable so that relationships such as Manning's equation can be used to describe resistance effects (to evaluate friction slope).
- The fluid is incompressible and of constant density throughout the flow.

Full equations

I. Continuity equation: Using the principle inflow-outflow = rate of change of storage

$$\frac{\partial Q}{\partial x} + \frac{\partial A}{\partial t} = q$$

II. Momentum equation: Based on Newton's second law

Net force = rate of change of momentum

$$\frac{\partial Q}{\partial t} + \frac{\partial(\beta Q^2/A)}{\partial x} + gA \left(\frac{\partial y}{\partial x} + S_f - S_0 + S_e \right) + W_f b - \beta q v_x = 0$$

where Q = inflow discharge; A = cross-sectional area of flow; q = lateral flow per unit length along the channel; β = momentum coefficient (or correction factor); S_f = friction slope; S_0 = bed slope; S_e = eddy loss slope; W_f = wind shear force; b = channel top width; v_x = velocity of lateral flow in the direction of channel flow

Simplified and most widely used form of St. Venant Equations (conservation forms)

The most common form of equation (taking $\beta = 1$; neglecting eddy loss, wind shear and lateral flow)

$$\text{Continuity: } \frac{\partial Q}{\partial x} + \frac{\partial A}{\partial t} = 0 \quad (\text{I})$$

$$\text{Momentum: } \frac{\partial Q}{\partial t} + \frac{\partial (Q^2/A)}{\partial x} + gA \frac{\partial y}{\partial x} - gA(S_0 - S_f) = 0 \quad (\text{II})$$

Terms in momentum equation

Effect of inertia force: first and second terms

First term: local acceleration

Second term: convective acceleration

Third term: pressure force

Fourth term: gravity force

Fifth term: friction force

In the Saint Venant equations

- All the flow variables are functions of both time and distance along the channel.
- The unknowns are Q and y in conservative forms of equations, and V and y in non-conservation form.
- The other flow variables, such as the area (A), and the friction slope (S_f) and can be expressed in terms of Q and y .
- The independent variables are time (t), and distance along the channel (x).
- An initial condition and two boundary conditions are needed to solve the Saint Venant equations.
- The initial condition is described by the variation of the unknowns, Q and y (or V and y), along the channel at time zero.

Simplifications of St. Venant equations

a. Dynamic wave model:

Full momentum equation is considered in dynamic wave model.

Equations (I) and (II) represent equations for dynamic wave model.

$$\text{Continuity: } \frac{\partial Q}{\partial x} + \frac{\partial A}{\partial t} = 0 \quad (\text{I})$$

$$\text{Momentum: } \frac{\partial Q}{\partial t} + \frac{\partial (Q^2/A)}{\partial x} + gA \frac{\partial y}{\partial x} - gA(S_0 - S_f) = 0 \quad (\text{II})$$

Application

- Flood wave in a wide river
- For rivers in which flood is modified by developments and structures (i.e., mild slope channels with downstream control)

b. Diffusive wave model:

Neglecting first two terms (inertia) of momentum equation, diffusive wave equation is obtained. Acceleration terms are small compared to water surface slope for slow rising of flood wave, which can be neglected

$$\text{Continuity: } \frac{\partial Q}{\partial x} + \frac{\partial A}{\partial t} = 0$$

$$\text{Momentum: } \frac{\partial y}{\partial x} = S_0 - S_f \quad (\text{III})$$

Application

- To represent backwater effect
- Steep slope channel with flow influenced by bed roughness

c. Kinematic wave model

Neglecting first three terms of momentum equation, kinematic wave equation is obtained.

$$\text{Continuity: } \frac{\partial Q}{\partial x} + \frac{\partial A}{\partial t} = q \quad (\text{I})$$

$$\text{Momentum: } S_0 = S_f \quad (\text{II})$$

In this model, gravity force and friction force balance each other. The wave motion is described principally by the equation of continuity. No force term is included (kinematic). So, it is equivalent to uniform flow (momentum equation) i.e., flow remaining approximately uniform over a channel reach, and no visible surface wave.

Application

- Steep slope channels with no back water effects,
- Overland flow (low depth of flow, bed slope approximately equal to friction slope) where lateral flow is continuously added,
- Flow from small basins

Momentum equation in kinematic wave model can be expressed as

$$A = \alpha Q^\beta \quad (\text{III})$$

Differentiating equation (III) with respect to t

$$\frac{\partial A}{\partial t} = \alpha \beta Q^{\beta-1} \frac{\partial Q}{\partial t} \quad (\text{IV})$$

From equations (I) and (IV)

$$\frac{\partial Q}{\partial x} + \alpha \beta Q^{\beta-1} \frac{\partial Q}{\partial t} = q \quad (\text{V})$$

This is the kinematic wave model in terms of Q (single equation).

Using Manning's equation

$$Q = \frac{1}{n} A R^{2/3} S^{1/2} = \frac{1}{n} \frac{A^{5/3}}{P^{2/3}} S^{1/2}$$

$$\text{i.e., } A = \left(\frac{n P^{2/3}}{\sqrt{S}} \right)^{3/5} Q^{3/5} \quad (\text{VI})$$

Comparing it to equation (III)

$$\alpha = \left(\frac{n P^{2/3}}{\sqrt{S}} \right)^{0.6} \text{ and } \beta = 0.6$$

Saint Venant equation in terms of V and y (non-conservation form)

$$\text{Continuity: } \frac{\partial Q}{\partial x} + \frac{\partial A}{\partial t} = 0 \quad (\text{Basic form})$$

$$\text{or } \frac{\partial(AV)}{\partial x} + \frac{\partial A}{\partial y} \frac{\partial y}{\partial t} = 0$$

$$\text{or } A \frac{\partial V}{\partial x} + V \frac{\partial A}{\partial x} + T \frac{\partial y}{\partial t} = 0$$

$$\text{or } A \frac{\partial V}{\partial x} + V \frac{T \partial y}{\partial x} + T \frac{\partial y}{\partial t} = 0$$

Dividing throughout by T

$$D \frac{\partial V}{\partial x} + V \frac{\partial y}{\partial x} + \frac{\partial y}{\partial t} = 0 \quad (\text{where } D=A/T = \text{hydraulic depth})$$

$$\text{i.e., } \frac{\partial y}{\partial t} + V \frac{\partial y}{\partial x} + D \frac{\partial V}{\partial x} = 0$$

$$\text{Momentum: } \frac{\partial Q}{\partial t} + \frac{\partial(Q^2/A)}{\partial x} + gA \frac{\partial y}{\partial x} - gA(S_0 - S_f) = 0$$

$$\text{or, } \frac{\partial(AV)}{\partial t} + \frac{\partial(QV)}{\partial x} + gA \frac{\partial y}{\partial x} - gA(S_0 - S_f) = 0$$

$$\text{or, } A \frac{\partial V}{\partial t} + V \frac{\partial A}{\partial t} + V \frac{\partial Q}{\partial x} + Q \frac{\partial V}{\partial x} + gA \frac{\partial y}{\partial x} - gA(S_0 - S_f) = 0$$

$$\text{or, } A \frac{\partial V}{\partial t} + V \left(\frac{\partial A}{\partial t} + \frac{\partial Q}{\partial x} \right) + AV \frac{\partial V}{\partial x} + gA \frac{\partial y}{\partial x} - gA(S_0 - S_f) = 0$$

Substituting $\frac{\partial Q}{\partial x} + \frac{\partial A}{\partial t} = 0$ (from continuity) and dividing throughout by A

$$\frac{\partial V}{\partial t} + V \frac{\partial V}{\partial x} + g \frac{\partial y}{\partial x} - g(S_0 - S_f) = 0$$

5.4 Numerical scheme (FDM) for kinematic wave model

For numerical solution of kinematic wave equation, channel properties, initial condition (value of variable, e.g. Q at time $t = 0$) and boundary conditions (value of variable at two ends of a channel reach) should be given.

The FDM can be formulated in either linear or non-linear form.

In linear scheme, Kinematic equation in terms of Q is used.

In non-linear scheme, original continuity and alternative form of momentum equation are used.

A. Linear scheme for kinematic wave model

(I) First order accurate implicit scheme

Equation to solve: $\frac{\partial Q}{\partial x} + \alpha \beta Q^{\beta-1} \frac{\partial Q}{\partial t} = q$

To find: Q_{i+1}^{n+1}

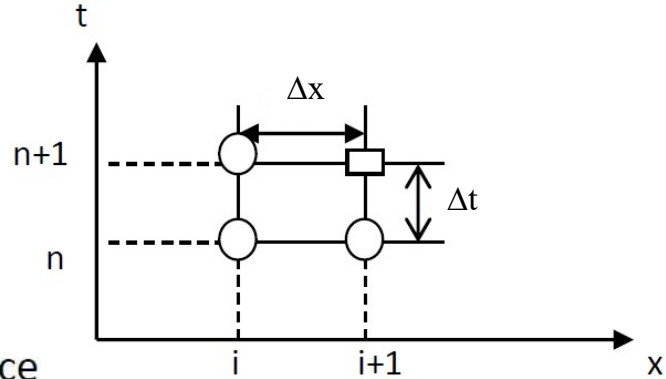
○ : known points

□ : computation point

Discretization of derivative terms

$$\frac{\partial Q}{\partial x} = \frac{Q_{i+1}^{n+1} - Q_i^{n+1}}{\Delta x} : \text{backward difference}$$

$$\frac{\partial Q}{\partial t} = \frac{Q_{i+1}^{n+1} - Q_{i+1}^n}{\Delta t} : \text{backward difference}$$



Discretization of non-derivative terms

$$Q = \frac{Q_{i+1}^n + Q_i^{n+1}}{2} : \text{average along diagonal for linearity}$$

$$q = \frac{q_{i+1}^{n+1} + q_{i+1}^n}{2} : \text{average on } i+1 \text{ distance line (q values to be given for all grids)}$$

Substituting above values in $\frac{\partial Q}{\partial x} + \alpha \beta Q^{\beta-1} \frac{\partial Q}{\partial t} = q$

$$\frac{Q_{i+1}^{n+1} - Q_i^{n+1}}{\Delta x} + \alpha \beta \left(\frac{Q_{i+1}^n + Q_i^{n+1}}{2} \right)^{\beta-1} \frac{Q_{i+1}^{n+1} - Q_{i+1}^n}{\Delta t} = \frac{q_{i+1}^{n+1} + q_{i+1}^n}{2}$$

$$\text{or, } Q_{i+1}^{n+1} \left(\frac{\Delta t}{\Delta x} + \alpha \beta \left(\frac{Q_{i+1}^n + Q_i^{n+1}}{2} \right)^{\beta-1} \right) = \frac{\Delta t}{\Delta x} Q_i^{n+1} + \alpha \beta Q_{i+1}^n \left(\frac{Q_{i+1}^n + Q_i^{n+1}}{2} \right)^{\beta-1} + \left(\frac{q_{i+1}^{n+1} + q_{i+1}^n}{2} \right) \Delta t$$

$$\text{i.e., } Q_{i+1}^{n+1} = \frac{\frac{\Delta t}{\Delta x} Q_i^{n+1} + \alpha \beta Q_{i+1}^n \left(\frac{Q_{i+1}^n + Q_i^{n+1}}{2} \right)^{\beta-1} + \left(\frac{q_{i+1}^{n+1} + q_i^n}{2} \right) \Delta t}{\left(\frac{\Delta t}{\Delta x} + \alpha \beta \left(\frac{Q_{i+1}^n + Q_i^{n+1}}{2} \right)^{\beta-1} \right)} \quad (\text{a})$$

Solution technique for linear kinematic wave model for channel routing (Algorithm)

1. Supply initial condition at time $t = 0$ on time line $n = 1$.
2. Apply Courant condition to each grid point on time line and choose the smallest value as Δt .
3. Advance to next time step, $t = t + \Delta t$ and $n = n + 1$.
 - 3.1 Apply upstream boundary conditions (flow hydrograph) to obtain Q_1^{n+1} at $i=1$.

3.2 Move to next point on time line $n+1$ by Δx ($x = x + \Delta x$).

3.3 Solve for Q_{i+1}^{n+1} using equation (a).

3.4 Repeat steps 3.2-3.3 for all points on the grid.

4. Repeat steps 3.1-3.4 for the remaining time periods.

For explicit scheme, discretize $\frac{\partial Q}{\partial x}$ at n time level, i.e. Replace $\frac{\partial Q}{\partial x}$ by $\frac{Q_{i+1}^n - Q_{i-1}^n}{2\Delta x}$. All other terms are similar to implicit scheme.

(II) Second order accurate scheme (Central difference scheme: explicit)

Equation to solve: $\frac{\partial Q}{\partial x} + \alpha\beta Q^{\beta-1} \frac{\partial Q}{\partial t} = q$

To find: Q_i^{n+1}

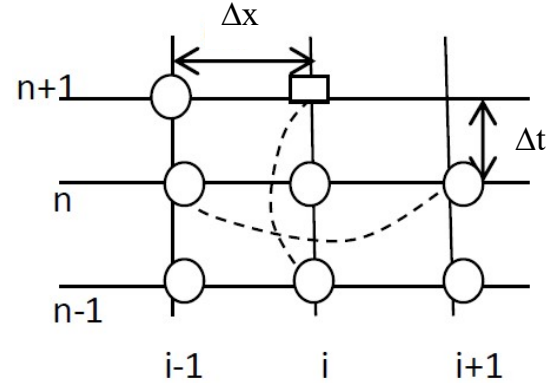
○ : known points

□ : computation point

Discretization of derivative terms

$$\frac{\partial Q}{\partial x} = \frac{Q_{i+1}^n - Q_{i-1}^n}{2\Delta x} : \text{central difference}$$

$$\frac{\partial Q}{\partial t} = \frac{Q_i^{n+1} - Q_i^{n-1}}{2\Delta t} : \text{central difference}$$



Discretization of non-derivative terms

$$Q = \frac{Q_{i+1}^{n-1} + Q_i^n + Q_{i-1}^{n+1}}{3} : \text{average along diagonal for linearity}$$

$$q = \frac{q_i^{n+1} + q_i^n + q_i^{n-1}}{3} : \text{average on } i \text{ distance line (} q \text{ values to be given for all grids)}$$

Substituting above values in $\frac{\partial Q}{\partial x} + \alpha\beta Q^{\beta-1} \frac{\partial Q}{\partial t} = q$

$$\frac{Q_{i+1}^n - Q_{i-1}^n}{2\Delta x} + \alpha\beta \left(\frac{Q_{i+1}^{n-1} + Q_i^n + Q_{i-1}^{n+1}}{3} \right)^{\beta-1} \frac{Q_i^{n+1} - Q_i^{n-1}}{2\Delta t} = \frac{q_i^{n+1} + q_i^n + q_i^{n-1}}{3}$$

or, $Q_i^{n+1} \left(\alpha\beta \left(\frac{Q_{i+1}^{n-1} + Q_i^n + Q_{i-1}^{n+1}}{3} \right)^{\beta-1} \right) =$

$$-\frac{\Delta t}{\Delta x} (Q_{i+1}^n - Q_{i-1}^n) + \alpha\beta Q_i^{n-1} \left(\frac{Q_{i+1}^{n-1} + Q_i^n + Q_{i-1}^{n+1}}{3} \right)^{\beta-1} + \left(\frac{q_i^{n+1} + q_i^n + q_i^{n-1}}{3} \right) 2\Delta t$$

Q_i^{n+1} a = b : linear scheme where a = bracketed term in LHS, b = RHS=known

B. Non-linear scheme for kinematic wave model

Same discretization scheme as linear model is followed for space and time derivative.

The difference is that the continuity equation together with relationship of A and Q are used in the formulation of FDM equation for obtaining non-linear algebraic equation.

(I) First order accurate implicit scheme

Equation to solve: $\frac{\partial Q}{\partial x} + \frac{\partial A}{\partial t} = q$ (I)

To find: Q_{i+1}^{n+1}

○ : known points

□ : computation point

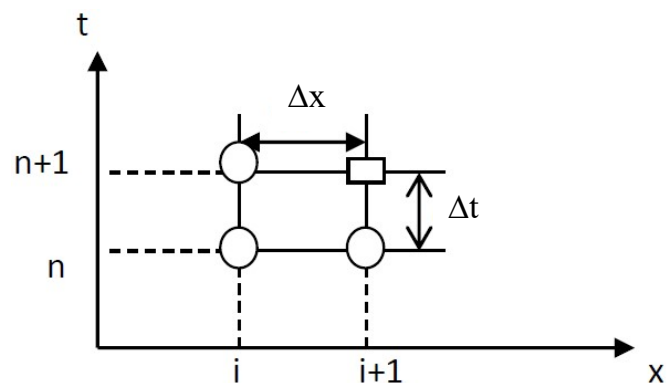
Discretization of derivative terms

$$\frac{\partial Q}{\partial x} = \frac{Q_{i+1}^{n+1} - Q_i^{n+1}}{\Delta x} : \text{backward}$$

$$\frac{\partial A}{\partial t} = \frac{A_{i+1}^{n+1} - A_{i+1}^n}{\Delta t} : \text{backward}$$

Discretization of non-derivative terms

$$q = \frac{q_{i+1}^{n+1} + q_{i+1}^n}{2} : \text{average on } i+1 \text{ distance line (q values to be given for all grids)}$$



Substituting in equation (I)

$$\frac{Q_{i+1}^{n+1} - Q_i^{n+1}}{\Delta x} + \frac{A_{i+1}^{n+1} - A_{i+1}^n}{\Delta t} = \frac{q_{i+1}^{n+1} + q_{i+1}^n}{2} \quad (\text{II})$$

With momentum equation $A = \alpha Q^\beta$

$$A_{i+1}^{n+1} = \alpha (Q_{i+1}^{n+1})^\beta \text{ and } A_{i+1}^n = \alpha (Q_{i+1}^n)^\beta$$

Substituting in equation (II)

$$\frac{Q_{i+1}^{n+1} - Q_i^{n+1}}{\Delta x} + \frac{\alpha (Q_{i+1}^{n+1})^\beta - \alpha (Q_{i+1}^n)^\beta}{\Delta t} = \frac{q_{i+1}^{n+1} + q_{i+1}^n}{2}$$

Solving for Q_{i+1}^{n+1}

$$\frac{\Delta t}{\Delta x} Q_{i+1}^{n+1} + \alpha (Q_{i+1}^{n+1})^\beta = \frac{\Delta t}{\Delta x} Q_i^{n+1} + \alpha (Q_{i+1}^n)^\beta + \Delta t \left(\frac{q_{i+1}^{n+1} + q_{i+1}^n}{2} \right) \quad (\text{III})$$

It is non-linear in Q_{i+1}^{n+1} and can be solved by iteration technique, such as Newton-Raphson method.

Let $\text{RHS} = c$ which is known.

$$\text{Residual error is } f(Q_{i+1}^{n+1}) = \frac{\Delta t}{\Delta x} Q_{i+1}^{n+1} + \alpha (Q_{i+1}^{n+1})^\beta - c \quad (\text{IV})$$

$$\text{i.e., } f'(Q_{i+1}^{n+1}) = \frac{\Delta t}{\Delta x} + \alpha \beta (Q_{i+1}^{n+1})^{\beta-1} \quad (\text{V})$$

The objective is to find Q_{i+1}^{n+1} that forces $f(Q_{i+1}^{n+1})$ to become zero.

From Newton-Raphson formula

$$(Q_{i+1}^{n+1})_{k+1} = (Q_{i+1}^{n+1})_k - \frac{f(Q_{i+1}^{n+1})_k}{f'(Q_{i+1}^{n+1})_k} \quad (\text{VI})$$

Any arbitrary value can be taken as initial value. However, for faster convergence, it is better to use Q_{i+1}^{n+1} obtained from linear scheme as initial condition.

Solution technique for non-linear kinematic wave model for channel routing (Algorithm)

1. Supply initial condition at time $t = 0$ on time line $n = 1$.
2. Advance to next time step, $t = t + \Delta t$ and $n = n + 1$.
 - 2.1 Apply upstream boundary conditions (flow hydrograph) to obtain Q_1^{n+1} at $i = 1$.
 - 2.2 Move to next point on time line $n + 1$ by Δx ($x = x + \Delta x$).
 - 2.3 Solve for initial estimate of Q_{i+1}^{n+1} at $k = 1$ using linear scheme. Find $f(Q_{i+1}^{n+1})_k$ using equation (IV).
 - 2.4 Find $f'(Q_{i+1}^{n+1})_k$ from equation (V).
 - 2.5 Find $(Q_{i+1}^{n+1})_{k+1}$ from equation (VI).
 - 2.6 Find $f(Q_{i+1}^{n+1})_{k+1}$ from equation (IV).
 - 2.7 If $f(Q_{i+1}^{n+1})_{k+1} \leq \varepsilon$, move to next grid point. Else repeat steps 2.4-2.6 until the condition is satisfied.
 - 2.8 Repeat steps 2.2-2.7 until the last grid point is reached.
3. Repeat steps 2.1-2.8 for the remaining time periods.

(II) Second order accurate scheme (Central difference scheme: explicit)

Equation to solve: $\frac{\partial Q}{\partial x} + \frac{\partial A}{\partial t} = q$ (I)

To find: Q_i^{n+1}

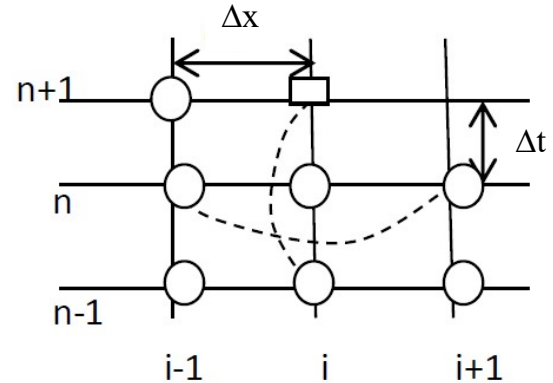
○ : known points

□ : computation point

Discretization of derivative terms

$$\frac{\partial Q}{\partial x} = \frac{Q_{i+1}^n - Q_{i-1}^n}{2\Delta x} : \text{central difference}$$

$$\frac{\partial A}{\partial t} = \frac{A_i^{n+1} - A_i^{n-1}}{2\Delta t} : \text{central difference}$$



Discretization of non-derivative terms

$$q = \frac{q_i^{n+1} + q_i^n + q_i^{n-1}}{3} : \text{average on } i \text{ distance line (} q \text{ values to be given for all grids)}$$

Substituting above values in equation (I)

$$\frac{Q_{i+1}^n - Q_{i-1}^n}{2\Delta x} + \frac{A_i^{n+1} - A_i^{n-1}}{2\Delta t} = \frac{q_i^{n+1} + q_i^n + q_i^{n-1}}{3} \quad \text{(II)}$$

With momentum equation $A = \alpha Q^\beta$

$$A_i^{n+1} = \alpha (Q_i^{n+1})^\beta \text{ and } A_i^{n-1} = \alpha (Q_i^{n-1})^\beta$$

Substituting in equation (II)

$$\frac{Q_{i+1}^n - Q_{i-1}^n}{2\Delta x} + \frac{\alpha(Q_i^{n+1})^\beta - \alpha(Q_i^{n-1})^\beta}{2\Delta t} = \frac{q_i^{n+1} + q_i^n + q_i^{n-1}}{3}$$

Solving for Q_i^{n+1}

$$\alpha(Q_i^{n+1})^\beta = -\frac{\Delta t}{\Delta x}(Q_{i+1}^n - Q_{i-1}^n) + \alpha(Q_i^{n-1})^\beta + 2\Delta t\left(\frac{q_i^{n+1} + q_i^n + q_i^{n-1}}{3}\right) \quad (\text{III})$$

It is non-linear in Q_i^{n+1} ; and can be solved directly.

Stability of numerical solution of kinematic wave model (explicit scheme)

A necessary condition for explicit scheme to be stable is the Courant condition. Courant number is a ratio of actual wave speed to numerical wave speed.

$$\text{Courant number} = \frac{V + |c|}{\Delta x / \Delta t}$$

where V = flow velocity, and c = celerity of wave.

The relative velocity of a wave with respect to the speed of fluid is called celerity.

Celerity of dynamic wave, $c = \sqrt{g y}$ for rectangular channel
where y = depth of flow.

For other channels, $c = \sqrt{g A/T}$

where A = cross-sectional area, and T = top width

Courant number ≤ 1 for stability

For computational purposes in an explicit scheme, Δx is specified and kept fixed throughout the computations, while Δt is determined at each time step. Δt at each point on a time line is computed satisfying Courant condition and the smallest one is used.

Stability of kinematic wave (explicit scheme)

Neglecting V (small compared to c),

$$\text{Courant number} = \frac{C_k}{\Delta x / \Delta t}$$

where C_k = kinematic wave celerity

Courant condition:

$$\frac{C_k}{\Delta x / \Delta t} \leq 1$$

$$\text{i.e., } \Delta t \leq \Delta x / C_k$$

C_k is given as

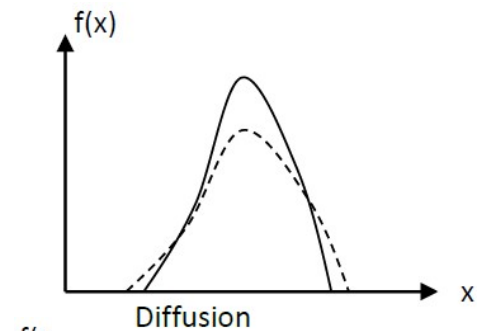
$$C_k = \frac{dQ}{dA} = \frac{1}{b} \frac{dQ}{dy}$$

where b = width of channel

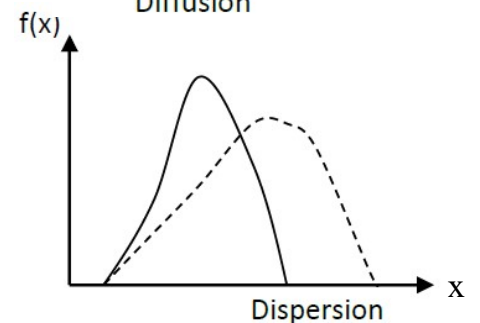
Apply Courant condition to each grid point on time line and choose the smallest value of time step.

Diffusion and Dispersion in the solution of kinematic wave equations by FDM

Diffusion: Numerical damping caused by the truncation error is called diffusion. It is related to amplitude error, in which the peak computed by numerical scheme reduces but the timing of peak is same and the shape of function becomes wide.



Dispersion: Error in timing of peak in the numerical solution is called dispersion. It is related to phase error, in which timing of the peak computed by numerical scheme is shifted from the real solution.



5.5 Finite difference scheme for Saint Venant equations (Dynamic wave model)

A. Solution by explicit scheme

(I) Example of first order accurate scheme

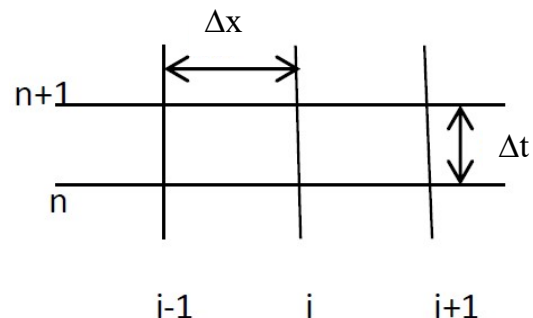
Forward difference for space and time derivatives

$$\frac{\partial f}{\partial x} = \frac{f_{i+1}^n - f_i^n}{\Delta x}$$

$$\frac{\partial f}{\partial t} = \frac{f_i^{n+1} - f_i^n}{\Delta t}$$

Continuity equation

$$\frac{\partial Q}{\partial x} + \frac{\partial A}{\partial t} = 0$$



$$\text{i.e., } \frac{A_i^{n+1} - A_i^n}{\Delta t} + \frac{Q_{i+1}^n - Q_i^n}{\Delta x} = 0$$

$$\text{i.e., } A_i^{n+1} = A_i^n - \Delta t \frac{Q_{i+1}^n - Q_i^n}{\Delta x}$$

Momentum equation

$$\frac{\partial Q}{\partial t} + \frac{\partial (Q^2/A)}{\partial x} + gA \frac{\partial y}{\partial x} - gA(S_0 - S_f) = 0$$

$$\text{i.e., } \frac{Q_i^{n+1} - Q_i^n}{\Delta t} + \frac{1}{\Delta x} [(Q^2/A)_{i+1}^n - (Q^2/A)_i^n] + gA_i^n \frac{y_{i+1}^n - y_i^n}{\Delta x} - gA_i^n (S_0 - S_f)_i^n = 0$$

$$\text{i.e., } Q_i^{n+1} = Q_i^n - \Delta t \left[\frac{1}{\Delta x} \{ (Q^2/A)_{i+1}^n - (Q^2/A)_i^n \} + gA_i^n \frac{y_{i+1}^n - y_i^n}{\Delta x} - gA_i^n (S_0 - S_f)_i^n \right]$$

(II) Example of second order accurate scheme (Leap-frog scheme or central difference scheme)

Central difference for space and time derivatives

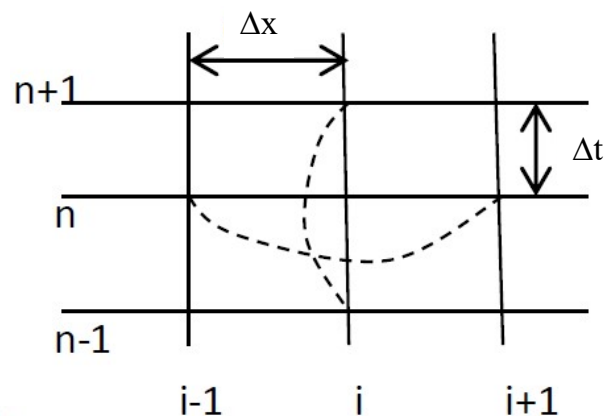
Discretization of derivative terms

$$\frac{\partial f}{\partial x} = \frac{f_{i+1}^n - f_{i-1}^n}{2\Delta x}$$

$$\frac{\partial f}{\partial t} = \frac{f_i^{n+1} - f_i^{n-1}}{2\Delta t}$$

Discretization of non-derivative terms

$$f = f_i^n \text{ or } \frac{f_{i-1}^n + f_{i+1}^n}{2} \text{ or } \frac{f_{i-1}^n + f_i^n + f_{i+1}^n}{3}$$



Continuity equation

$$\frac{\partial Q}{\partial x} + \frac{\partial A}{\partial t} = 0$$

$$\text{i.e., } \frac{A_i^{n+1} - A_i^{n-1}}{2\Delta t} + \frac{Q_{i+1}^n - Q_{i-1}^n}{2\Delta X} = 0$$

$$\text{i.e., } A_i^{n+1} = A_i^{n-1} - \Delta t \frac{Q_{i+1}^n - Q_{i-1}^n}{\Delta X}$$

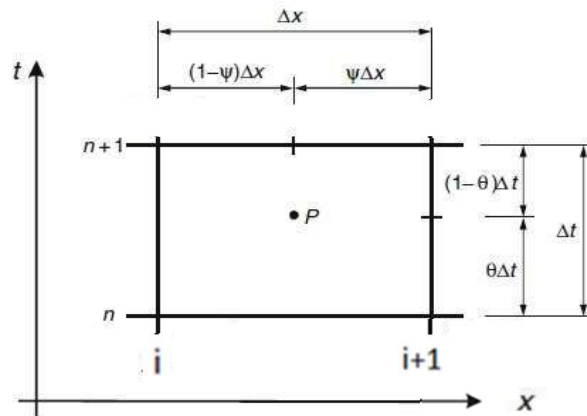
Momentum equation

$$\frac{\partial Q}{\partial t} + \frac{\partial(Q^2/A)}{\partial x} + gA \frac{\partial y}{\partial x} - gA(S_0 - S_f) = 0$$

$$\text{i.e., } \frac{Q_{i+1}^{n+1} - Q_{i-1}^{n-1}}{2\Delta t} + \frac{1}{2\Delta x} [(Q^2/A)_{i+1}^n - (Q^2/A)_{i-1}^n] + gA_i^n \frac{y_{i+1}^n - y_{i-1}^n}{2\Delta X} - gA_i^n (S_0 - S_f)_i^n = 0$$

$$\text{i.e., } Q_i^{n+1} = Q_i^{n-1} - 2\Delta t \left[\frac{1}{2\Delta x} \{ (Q^2/A)_{i+1}^n - (Q^2/A)_{i-1}^n \} + gA_i^n \frac{y_{i+1}^n - y_{i-1}^n}{2\Delta X} - gA_i^n (S_0 - S_f)_i^n \right]$$

B. Solution by implicit scheme



$$\frac{\partial f}{\partial x} = \theta \frac{(f_{i+1}^{n+1} - f_i^{n+1})}{\Delta X} + (1 - \theta) \frac{(f_{i+1}^n - f_i^n)}{\Delta X}$$

$$\frac{\partial f}{\partial t} = \psi \frac{(f_i^{n+1} - f_i^n)}{\Delta t} + (1 - \psi) \frac{(f_{i+1}^{n+1} - f_{i+1}^n)}{\Delta t}$$

B. Solution by implicit scheme

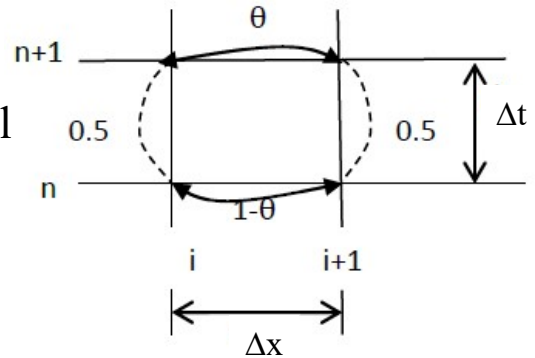
Implicit four point scheme (Preissman scheme or box scheme)

In Preissmann scheme, four points are used to evaluate the derivatives. The scheme is forward difference scheme with weight.

Discretization of derivative terms

$$\frac{\partial f}{\partial x} = \theta \frac{(f_{i+1}^{n+1} - f_i^{n+1})}{\Delta X} + (1 - \theta) \frac{(f_{i+1}^n - f_i^n)}{\Delta X}$$

$$\frac{\partial f}{\partial t} = \frac{1}{2} \left(\frac{f_{i+1}^{n+1} - f_{i+1}^n}{\Delta t} + \frac{f_i^{n+1} - f_i^n}{\Delta t} \right) \text{ (Giving equal weightage)}$$



Discretization of non-derivative terms

$$f = \frac{\theta}{2} (f_{i+1}^{n+1} + f_i^{n+1}) + \frac{(1-\theta)}{2} (f_{i+1}^n + f_i^n)$$

$\theta = 0.5$: box scheme

$\theta = 0$: fully explicit

$\theta = 1$: fully implicit

For implicit scheme, θ should be between 0.5 to 1.

For θ in between 0.5 to 1, the scheme is unconditionally stable.

Continuity equation

$$\frac{\partial Q}{\partial x} + \frac{\partial A}{\partial t} = 0$$

$$\text{i.e., } \frac{1}{2} \left(\frac{A_{i+1}^{n+1} - A_{i+1}^n}{\Delta t} + \frac{A_i^{n+1} - A_i^n}{\Delta t} \right) + \theta \frac{(Q_{i+1}^{n+1} - Q_i^{n+1})}{\Delta X} + (1 - \theta) \frac{(Q_{i+1}^n - Q_i^n)}{\Delta X} = 0 \quad (1)$$

Momentum equation

$$\frac{\partial Q}{\partial t} + \frac{\partial(Q^2/A)}{\partial x} + gA \frac{\partial y}{\partial x} - gA(S_0 - S_f) = 0$$

$$\begin{aligned} \text{i.e., } \frac{1}{2} \left(\frac{Q_{i+1}^{n+1} - Q_{i+1}^n}{\Delta t} + \frac{Q_i^{n+1} - Q_i^n}{\Delta t} \right) + \theta \frac{\{(Q^2/A)_{i+1}^{n+1} - (Q^2/A)_i^{n+1}\}}{\Delta X} + (1-\theta) \frac{\{(Q^2/A)_{i+1}^n - (Q^2/A)_i^n\}}{\Delta X} \\ + g \left\{ \frac{\theta}{2} (A_{i+1}^{n+1} + A_i^{n+1}) + \frac{(1-\theta)}{2} (A_{i+1}^n + A_i^n) \right\} \left\{ \theta \frac{(y_{i+1}^{n+1} - y_i^{n+1})}{\Delta X} + (1-\theta) \frac{(y_{i+1}^n - y_i^n)}{\Delta X} \right\} \\ - g \left\{ \frac{\theta}{2} (A_{i+1}^{n+1} + A_i^{n+1}) + \frac{(1-\theta)}{2} (A_{i+1}^n + A_i^n) \right\} \left\{ \frac{\theta}{2} ((S_0 - S_f)_{i+1}^{n+1} + (S_0 - S_f)_i^{n+1}) + \frac{(1-\theta)}{2} ((S_0 - S_f)_{i+1}^n + (S_0 - S_f)_i^n) \right\} = 0 \end{aligned} \quad (\text{II})$$

The unknown quantities in implicit finite difference equations are Q_i^{n+1} and y_i^{n+1} (the terms Q_{i+1}^{n+1} and y_{i+1}^{n+1} should also be computed simultaneously).

The area A_i^{n+1} and the friction slope $(S_f)_i^{n+1}$ can be expressed in terms of Q_i^{n+1} and y_i^{n+1} . All the other terms are known either from the initial conditions or from the previous time step computations.

For $i = 1, 2, \dots, N$, there are $N-1$ reaches. For each reach, 2 equations (continuity and momentum) can be written. So, there are $2N-2$ equations. Additional 2 equations are provided from boundary conditions.

The upstream boundary condition is usually specified as a known inflow hydrograph, while the downstream boundary condition can be specified as a known stage hydrograph, a known discharge hydrograph, or a known relationship between stage and discharge, such as a rating curve.

Solution technique of resulting equations:

The system of linear equations can be expressed in functional form in terms of the unknowns y and Q at time level $n+1$, as follows

$UB(y_1, Q_1) = 0$	upstream boundary condition
$C_1(y_1, Q_1, y_2, Q_2) = 0$	continuity for grid 1
$M_1(y_1, Q_1, y_2, Q_2) = 0$	momentum for grid 1
$\vdots$	
$C_i(y_i, Q_i, y_{i+1}, Q_{i+1}) = 0$	continuity for grid i
$M_i(y_i, Q_i, y_{i+1}, Q_{i+1}) = 0$	momentum for grid i
$\vdots$	
$C_{N-1}(y_{N-1}, Q_{N-1}, y_N, Q_N) = 0$	continuity for grid $N - 1$
$M_{N-1}(y_{N-1}, Q_{N-1}, y_N, Q_N) = 0$	momentum for grid $N - 1$
$DB(y_N, Q_N) = 0$	downstream boundary condition

This system of $2N$ nonlinear equations in $2N$ unknown is solved for each time step by Newton-Raphson method.

The computational procedure for each time $n+1$ starts by assigning trail values of $2N$ unknowns at that time.

These trail values of Q and y can be the values known at time n from the initial condition (if $n=1$) or from calculations during the previous time step.

Using trail values in the system of equations results in $2N$ residuals. For the k^{th} iteration these residuals can be expressed as

$UB(y_1^k, Q_1^k) = RUB^k$	residual for upstream boundary condition
$C_1(y_1^k, Q_1^k, y_2^k, Q_2^k) = RC_1^k$	residual for continuity at grid 1
$M_1(y_1^k, Q_1^k, y_2^k, Q_2^k) = RM_1^k$	residual for momentum at grid 1
$\vdots$	
$C_i(y_i^k, Q_i^k, y_{i+1}^k, Q_{i+1}^k) = RC_i^k$	residual for continuity at grid i
$M_i(y_i^k, Q_i^k, y_{i+1}^k, Q_{i+1}^k) = RM_i^k$	residual for momentum at grid i
$\vdots$	
$C_{N-1}(y_{N-1}^k, Q_{N-1}^k, y_N^k, Q_N^k) = RC_{N-1}^k$	residual for continuity at grid $N - 1$
$M_{N-1}(y_{N-1}^k, Q_{N-1}^k, y_N^k, Q_N^k) = RM_{N-1}^k$	residual for momentum at grid $N - 1$
$DB(y_N^k, Q_N^k) = RDB^k$	residual for downstream boundary condition

The solution is approached by finding values of the unknowns Q and y so that the residuals are forced to zero or very close to zero.

$$\text{i.e., } f(x) = 0$$

where $x = (Q_1, y_1, Q_2, y_2, \dots, Q_N, y_N)$ is linear vector of unknown quantities and for iteration k , $x^k = (Q_1^k, y_1^k, Q_2^k, y_2^k, \dots, Q_N^k, y_N^k)$. The nonlinear system can be linearized to

$$f(x^{k+1}) = f(x^k) + J(x^k)(x^{k+1} - x^k) = 0$$

where $J(x^k)$ is Jacobian, which is a coefficient matrix made up of the first partial derivatives of $f(x)$ evaluated at x^k .

The above equation can be written as

$$J(x^k)(x^{k+1} - x^k) = -f(x^k)$$

The system is solved for $(x^{k+1} - x^k) = \Delta x^k$, and the improved estimate of the solution, x^{k+1} , is determined knowing Δx^k . The process is repeated until $(x^{k+1} - x^k)$ is smaller than some specified tolerance.

The resulting system of equations is

$$\frac{\partial UB}{\partial y_1} dy_1 + \frac{\partial UB}{\partial Q_1} dQ_1 = -RUB^k$$

$$\frac{\partial C_1}{\partial y_1} dy_1 + \frac{\partial C_1}{\partial Q_1} dQ_1 + \frac{\partial C_1}{\partial y_2} dy_2 + \frac{\partial C_1}{\partial Q_2} dQ_2 = -RC_1^k$$

$$\frac{\partial M_1}{\partial y_1} dy_1 + \frac{\partial M_1}{\partial Q_1} dQ_1 + \frac{\partial M_1}{\partial y_2} dy_2 + \frac{\partial M_1}{\partial Q_2} dQ_2 = -RM_1^k$$

$$\vdots$$

$$\frac{\partial C_i}{\partial y_i} dy_i + \frac{\partial C_i}{\partial Q_i} dQ_i + \frac{\partial C_i}{\partial y_{i+1}} dy_{i+1} + \frac{\partial C_i}{\partial Q_{i+1}} dQ_{i+1} = -RC_i^k$$

$$\frac{\partial M_i}{\partial y_i} dy_i + \frac{\partial M_i}{\partial Q_i} dQ_i + \frac{\partial M_i}{\partial y_{i+1}} dy_{i+1} + \frac{\partial M_i}{\partial Q_{i+1}} dQ_{i+1} = -RM_i^k$$

$$\frac{\partial C_{N-1}}{\partial y_{N-1}} dy_{N-1} + \frac{\partial C_{N-1}}{\partial Q_{N-1}} dQ_{N-1} + \frac{\partial C_{N-1}}{\partial y_N} dy_N + \frac{\partial C_{N-1}}{\partial Q_N} dQ_N = -RC_{N-1}^k$$

$$\frac{\partial M_{N-1}}{\partial y_{N-1}} dy_{N-1} + \frac{\partial M_{N-1}}{\partial Q_{N-1}} dQ_{N-1} + \frac{\partial M_{N-1}}{\partial y_N} dy_N + \frac{\partial M_{N-1}}{\partial Q_N} dQ_N = -RM_{N-1}^k$$

$$\frac{\partial DB}{\partial y_N} dy_N + \frac{\partial DB}{\partial Q_N} dQ_N = -RDB^k$$

The Gaussian elimination or matrix inversion can be used to solve this set of equations.

Solution Algorithm

- Assign a set of trial values to the unknowns (Q and y) at all points. (Start with values of unknowns from initial condition, previous time step or from extrapolation).

- Put the values in LHS and compute residuals. Residual =(0-value of $f(x)$ in LHS).
- Find derivatives of continuity and momentum equations along all points w.r.t. Q and y . The derivatives form the matrix for $f'(x)$. dx part is correction dQ and dy . RHS = negative of residuals.
- Find the corrections by Matrix inversion.
- Apply the correction to previous value.
- Repeat the procedure until two values of corrections do not change significantly.
- Value at next time step = value at previous time step + correction.

Matrix form

Symbolic representation of continuity and momentum equation

$$C_i[Q_i^{n+1}, y_i^{n+1}, Q_{i+1}^{n+1}, y_{i+1}^{n+1}] = 0 \quad \text{Continuity equation}$$

$$M_i[Q_i^{n+1}, y_i^{n+1}, Q_{i+1}^{n+1}, y_{i+1}^{n+1}] = 0 \quad \text{Momentum equation}$$

Upstream boundary equation = B_I

Downstream boundary equation = B_N

Writing Equations for all the reaches , that is, for $i = 1$ to $(N-1)$

$$B_1[Q_1^{n+1}, y_1^{n+1}] = 0$$

$$C_1[Q_1^{n+1}, y_1^{n+1}, Q_2^{n+1}, y_2^{n+1}] = 0$$

$$M_1[Q_1^{n+1}, y_1^{n+1}, Q_2^{n+1}, y_2^{n+1}] = 0$$

$$C_2[Q_2^{n+1}, y_2^{n+1}, Q_3^{n+1}, y_3^{n+1}] = 0$$

$$M_2[Q_2^{n+1}, y_2^{n+1}, Q_3^{n+1}, y_3^{n+1}] = 0$$

.....

.....

$$C_{N-1}[Q_{N-1}^{n+1}, y_{N-1}^{n+1}, Q_N^{n+1}, y_N^{n+1}] = 0$$

$$M_{N-1}[Q_{N-1}^{n+1}, y_{N-1}^{n+1}, Q_N^{n+1}, y_N^{n+1}] = 0$$

$$B_N[Q_N^{n+1}, y_N^{n+1}] = 0$$

Residuals = $rB_1, rC_1, rM_1, rC_2, rM_2, \dots, rC_i, rM_i, \dots, rC_{N-1}, rM_{N-1}, rB_N$ (Residual = 0-value in LHS)

In Matrix form, $A \Delta = -F$

$$\begin{bmatrix} \frac{\partial B_1}{\partial Q_1} \frac{\partial B_1}{\partial y_1} \\ \frac{\partial C_1}{\partial Q_1} \frac{\partial C_1}{\partial y_1} \frac{\partial C_1}{\partial Q_2} \frac{\partial C_1}{\partial y_2} \\ \frac{\partial M_1}{\partial Q_1} \frac{\partial M_1}{\partial y_1} \frac{\partial M_1}{\partial Q_2} \frac{\partial M_1}{\partial y_2} \\ \frac{\partial C_2}{\partial Q_2} \frac{\partial C_2}{\partial y_2} \frac{\partial C_2}{\partial Q_3} \frac{\partial C_2}{\partial y_3} \\ \frac{\partial M_2}{\partial Q_2} \frac{\partial M_2}{\partial y_2} \frac{\partial M_2}{\partial Q_3} \frac{\partial M_2}{\partial y_3} \\ \vdots \\ \frac{\partial B_N}{\partial Q_N} \frac{\partial B_N}{\partial y_N} \end{bmatrix} \begin{bmatrix} \Delta Q_1 \\ \Delta y_1 \\ \Delta Q_2 \\ \Delta y_2 \\ \Delta Q_3 \\ \Delta y_3 \\ \vdots \\ \Delta Q_N \\ \Delta y_N \end{bmatrix} = \begin{bmatrix} -rB_1 \\ -rC_1 \\ -rM_1 \\ -rC_2 \\ -rM_2 \\ \vdots \\ -rB_N \end{bmatrix}$$

Remaining terms of coefficient matrix A are zero.

The solution of this system by any matrix inversion method, like the Gaussian elimination method, provides the corrections to the trial values of Q_i^{n+1} and y_i^{n+1} for the next iteration.

$$(Q_i^{n+1})_{k+1} = (Q_i^{n+1})_k + \Delta(Q_i)_k$$

$$(y_i^{n+1})_{k+1} = (y_i^{n+1})_k + \Delta(y_i)_k$$

This procedure is repeated until the corrections are reduced to tolerable magnitudes.

Boundary conditions

For subcritical flow: 2 boundary conditions, one at upstream and one at downstream

For supercritical flow: 2 boundary conditions, both at upstream

Types of boundary conditions

Discharge hydrograph

Stage hydrograph

Rating curve (stage-discharge relationship)

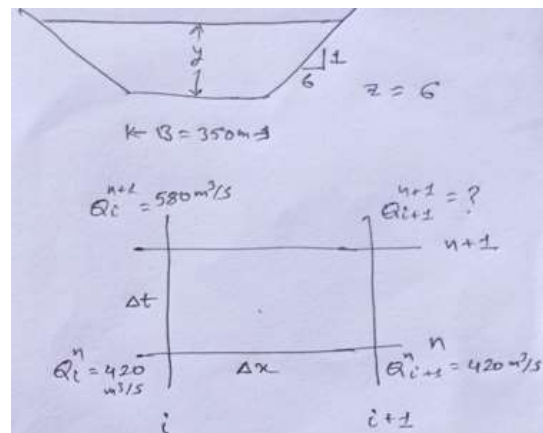
Normal depth

In a sub-critical flow a disturbance travels upstream as well downstream because the celerity is greater than the flow velocity (i.e., $v/c < 1$). Hence subcritical flows can be controlled from upstream and downstream as well (i.e., upstream and downstream flow control).

In a supercritical flow a disturbance cannot travel upstream because the celerity is less than the flow velocity (i.e., $v/c > 1$). Hence supercritical flows can only be controlled from upstream (i.e., upstream flow control). All rapid flows are controlled by the upstream flow conditions.

Problem: A river which can be generalized as a trapezoidal channel is 350 m wide with side slope 6:1 has a bed slope 1.5% and Manning's $n = 0.0359$. The initial discharge through the river is 420 cumecs. Due to a flood observed at upstream, the value of discharge rises to 580 cumecs. Calculate the discharge that will occur at 4.65 km downstream. Take $\Delta x = 4650$ m and $\Delta t = 1.5$ hours. Use non-linear kinematic wave solution.

$$\begin{aligned}
 S &= 1.5\% = 0.015, \quad n = 0.0359 \\
 Q &= 420 \text{ m}^3/\text{s}, \quad \Delta x = 4650 \text{ m}, \quad \Delta t = 1.5 \text{ hrs} \\
 &= 5400 \text{ sec.} \\
 Q &= \frac{1}{n} A \left(\frac{A}{P} \right)^{2/3} S^{1/2} \\
 &= \frac{1}{n} \frac{A^{5/3}}{P^{2/3}} S^{1/2} \\
 420 &= \frac{1}{0.0359} * \frac{(By + zy^2)^{5/3}}{(B + 2y\sqrt{1+z^2})^{2/3}} * \sqrt{0.015} \\
 \therefore 420 &= \frac{1}{0.0359} * \frac{(350y + 6y^2)^{5/3}}{(350 + 12.166y)^{2/3}} * \sqrt{0.015}
 \end{aligned}$$



Solving it, $d = 0.533 \text{ m}$

$$P = 350 + 12 \cdot 166 \cdot 0.533 = 356.484 \text{ m}$$

$$A = 350 \cdot 0.533 + 6 \cdot 0.533^2 = 188.255 \text{ m}^2$$

$$\alpha = \left(\frac{n P^{2/3}}{\sqrt{S}} \right)^{0.6} = \left(\frac{0.0359 \cdot 356.484^{2/3}}{\sqrt{0.015}} \right)^{0.6} = 5.024, \quad \beta = 0.6$$

for first trial

$$Q_{i+1}^{n+1} = \frac{\frac{\Delta t}{\Delta x} Q_i^{n+1} + \alpha \beta Q_{i+1}^n \left(\frac{Q_{i+1}^n + Q_i^{n+1}}{2} \right)^{\beta-1}}{\frac{\Delta t}{\Delta x} + \alpha \beta \left(\frac{Q_{i+1}^n + Q_i^{n+1}}{2} \right)^{\beta-1}}$$

$$= \frac{\frac{5400}{4650} \cdot 580 + 5.024 \cdot 0.6 \cdot 420 \left(\frac{420 + 580}{2} \right)^{0.6-1}}{\frac{5400}{4650} + 5.024 \cdot 0.6 \cdot \left(\frac{420 + 580}{2} \right)^{0.6-1}} = \frac{778.9536}{1.4123} = 551.55 \text{ m}^3/\text{s}$$

$$f(Q_{i+1}^{n+1}) = \frac{\Delta t}{\Delta x} Q_{i+1}^{n+1} + \alpha (Q_{i+1}^{n+1})^\beta - \left[\frac{\Delta t}{\Delta x} Q_i^{n+1} + \alpha (Q_{i+1}^n)^\beta \right]$$

$$= 1.16129 Q_{i+1}^{n+1} + 5.024 (Q_{i+1}^{n+1})^{0.6} - (1.16129 \cdot 580 + 5.024 \cdot 420^{0.6})$$

$$= 1.16129 Q_{i+1}^{n+1} + 5.024 (Q_{i+1}^{n+1})^{0.6} - 861.91295$$

$$f'(Q_{i+1}^{n+1}) = \frac{\Delta t}{\Delta x} + \alpha \beta (Q_{i+1}^{n+1})^{\beta-1} = 1.16129 + 5.024 \cdot 0.6 (Q_{i+1}^{n+1})^{0.6-1}$$

$$= 1.16129 + 3.0144 (Q_{i+1}^{n+1})^{-0.4}$$

$$(Q_{i+1}^{n+1})_{K+1} = (Q_{i+1}^{n+1})_K - \frac{f(Q_{i+1}^{n+1})_K}{f'(Q_{i+1}^{n+1})_K} = 551.55 - \frac{0.41648}{1.4026} = 551.253 \text{ m}^3/\text{s}$$

$$= 551.253 + \frac{9.894 \times 10^{-5}}{1.4026} = 551.253 \text{ m}^3/\text{s}$$